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//      CS 2130 - 9:30 am
//      Program #13
//      Dr. Huson
//      Due: 12/08/2022
//      Filename: 13Program.cpp
//      Version: 1.0
// -----
// makes a mealy machine to do cryptoquotes.
// -----

// Compiler directives
#include <iomanip>
#include <iostream>
using namespace std;

void readIn(char array[2][26]) {
    int i = 0;
    string inputStr;
    //fill the array with 0, priming it for reading in a new key
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 26; j++) {
            array[i][j] = '0';
        }
    }
    //i < 26 means until the mealy machine is finished being filled
    while (i < 26) {
        cout << "Please enter the key.  Charactes will be read until\n";
        cout << "26 unique characters are entered. For example:\n";
        cout << "mmnnb65247bvvcxzlkhgfdsapoiuy6trewq\n";
        cout << "would be accepted as the key: MNBVCXZLKJHGFDSAPOIUYTREWQ\n";
        cout << "Please enter a string of characters: ";
        cin >> inputStr;
        cin.ignore();
        for (int j = 0; j < inputStr.length(); j++) {
            if (isalpha(inputStr[j])) {
                inputStr[j] = toupper(inputStr[j]);
                if (array[1][inputStr[j] - 65] == '0') {
                    array[0][i] = inputStr[j];
                    array[1][inputStr[j] - 65] = i + 65;
                    i++;
                }
            }
        }
    }
    cout << endl;
}

void display(char array[2][26]) {
    //displays A-Z and then the encoding strip so you can follow the key yourself
    cout << "The key value is :\nA B C D E F G H I J K L M N O P Q R S T U V W X Y Z\n";
    for(int i = 0; i < 26; i++) {
        cout << array[0][i] << " ";
    }
    cout << endl;
}

void encode(char array[2][26]) {

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//uses the first half of the 2D array to encode their message
cout << "Please enter the string to create the Cryptoquote from as one line of
text.\nInput: ";
string inpStr;
getline(cin, inpStr);
//say they put in a C in for input the code'll read that and go and look what is in
//the third index of the encoding array and replace that character of the string
with it.
for(int i = 0; i < inpStr.length(); i++) {
    if(isalpha(inpStr[i])) {
        inpStr[i] = toupper(inpStr[i]);
        inpStr[i] = array[0][inpStr[i]-65];
    }
}
cout << "The resulting string is:\n" << inpStr << endl;
}

void decode(char array[2][26]) {
//uses the second half of the 2D array to decode their message
cout << "Please enter the Cryptoquote string to decode as one line of text.\n
Input: ";
string inpStr;
getline(cin, inpStr);
//say they put in an A in for input the code'll read that and go and look what is
in
//the first index of the decoding array and replace that character of the string
with it.
for(int i = 0; i < inpStr.length(); i++) {
    if(isalpha(inpStr[i])) {
        inpStr[i] = toupper(inpStr[i]);
        inpStr[i] = array[1][inpStr[i]-65];
    }
}
cout << "The resulting string is:\n" << inpStr << endl;
}

int main() {
char keyArray[2][26];
int input = 0;
while (input != 5) {
    cout << "Please choose from the following options:\n";
    cout << "\t1 - Enter a key\n";
    cout << "\t2 - Display a key\n";
    cout << "\t3 - Encode a quote\n";
    cout << "\t4 - Decode a Cryptoquote\n";
    cout << "\t5 - Exit program\n\n\tChoice? ";
    cin >> input;
    cin.ignore();
    switch (input) {
        case 1:
            readIn(keyArray);
            break;
        case 2:
            display(keyArray);
            break;
        case 3:
            encode(keyArray);
            break;
        case 4:

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        decode(keyArray);  
        break;  
    case 5:  
        cout << "Bye!!! Hope you solved your Cryptoquote!\n";  
        break;  
    }  
}  
}
```